

1] Consider a pump designed for incompressible liquids shown in the figure below. The size of the pump is characterized by the impeller diameter (d), it is assumed that all other dimension for the pump scale with this single length scale parameter. A motor is connected to the pump that spins the impeller at a constant angular velocity (Ω). In the test setup shown in the figure, the pump pulls in liquid of density ρ from the reservoir at pressure P_1 and the liquid discharges at the pump exit at pressure P_2 . A valve is used to adjust backpressure on the pump to simulate different flow resistances.

- Perform dimensional analysis following the steps presented in class to determine the non-dimensional relation between nondimensional pressure rise across the pump and the other variables. You can assume that the viscosity of the liquid can be neglected.
- Plot the data in the table below in non-dimensional form using the non-dimensional variables determined in your analysis.
- Does your analysis appear to have identified the correct non-dimensional relationship? Why?

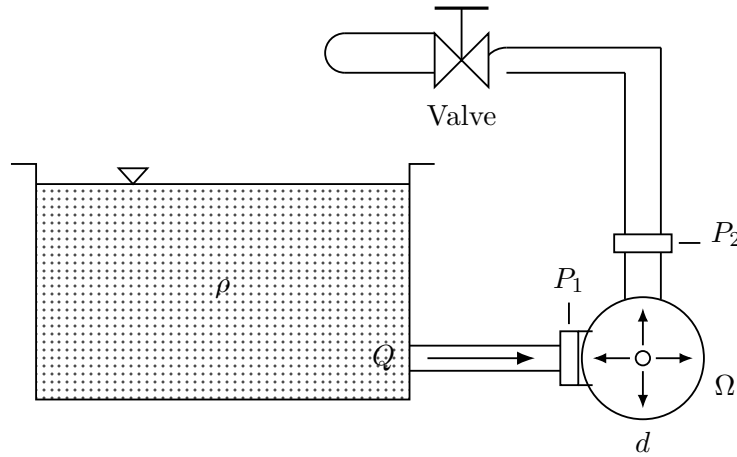


Table: Pump head ($H = \Delta P/(\rho g)$) generated as a function of flow rate (Q) at different pump speeds.

Speed = 1000 rpm		Speed = 1475 rpm		Speed = 1990 rpm	
Flowrate [g/min]	Head [ft]	Flowrate [g/min]	Head [ft]	Flowrate [g/min]	Head [ft]
0.0	30.4	0.0	64.7	0.0	113.4
23.5	29.7	12.6	63.6	9.3	113.1
38.7	29.0	33.8	62.4	22.7	111.0
51.0	28.6	65.1	60.3	36.5	110.9
74.8	27.0	96.4	57.3	59.9	110.1
87.8	25.2	115.8	56.0	81.9	109.0
101.2	24.3	139.9	53.8	107.5	105.0
120.6	22.8	149.6	54.2	126.9	105.2
		176.8	49.6	151.9	101.8
		188.0	47.2	173.4	100.1
				189.1	98.3
				208.1	95.9
				226.6	93.5
				240.0	90.6

2] A gas of non-interacting particles of mass m at temperature T has density ρ , and internal energy per unit volume ε .

- a) Using dimensional analysis, determine how ε must depend on ρ , T , and m . In your formulation use k_B = Boltzmann's constant, h = Planck's constant, and c = speed of light to include possible quantum and relativistic effects.
- b) Consider the limit of slow moving particles without quantum effects by requiring c and h to drop out of your dimensionless formulation. How does ε depend on ρ and T ? What type of gas follows this thermodynamic law?
- c) Consider the limit of massless particles (i.e. photons) by requiring m and ρ to drop out of your dimensionless formulation of part a). How does ε depend on T in this case? What is the name of this radiation law?

3] The vertical motion of a ship in water is governed by the z -component of the Navier–Stokes equation:

$$\frac{\partial w}{\partial t} + u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} = -\frac{1}{\rho} \frac{\partial p}{\partial z} - g + \frac{\mu}{\rho} \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} \right)$$

where u , v , w are the x , y , and z velocities respectively, and all other variables have their normal meanings. The flow velocity far from the ship is given by U and the length of the ship at the waterline is l .

- (a) Non-dimensionalize the governing equation, the z -component of Navier–Stokes.
- (b) What common non-dimensional parameters result?
- (c) Assume you didn't have knowledge of the governing equation but knew that the z -velocity, w , was a function of ρ , g , μ , U , l , find the non-dimensional parameters using the Buckingham PI theorem? (10 points)
- (d) We wish to test a scaled model of a tanker ship at scale of $\frac{1}{100}$ the actual size, is it practical to achieve dynamic similarity?